Primer: Sieve Estimation of Structural Models (SEES)

econirl package documentation

1 Problem and Motivation

The nested fixed point algorithm (NFXP) of Rust (1987) solves the Bellman equation to machine precision at every candidate parameter vector, costing $O(|\mathcal{S}|^2)$ per inner iteration. The CCP estimator of Hotz and Miller (1993) eliminates the inner loop but requires an $O(|\mathcal{S}|^3)$ matrix inversion. Both methods scale with the full state space. When $|\mathcal{S}|$ reaches hundreds or thousands of states, as in multi-component replacement models or empirical games, the computational cost becomes prohibitive.

Luo and Sang (2024) propose the Sieve-based Efficient Estimator for Structural models (SEES). The value function is approximated by a small set of deterministic basis functions, and the Bellman equation is imposed as a penalty rather than a constraint. This produces a single unconstrained optimization over the structural parameters θ and basis coefficients α, with computational cost $O(K)$ independent of $|\mathcal{S}|$. No inner loop is solved. No neural network is trained. When $K = |\mathcal{S}|$ and the penalty weight goes to infinity, SEES reduces to the MPEC estimator of Su and Judd (2012).

2 Model and Notation

Table 1: Common DDC notation shared across all primers.

Symbol	Name	Definition
$s \in \mathcal{S}$	State	Discrete or continuous
$a \in \mathcal{A}$	Action	Finite: $\lvert\mathcal{A}\rvert = J$
s'	Next state	Drawn from $P(s' \mid s, a)$
$\beta \in [0, 1)$	Discount factor	
$\sigma > 0$	Scale parameter	Logit noise scale ($= 1$ by convention)
$\theta \in \Theta \subset \mathbb{R}^{d_\theta}$	Structural parameters	Unknown
$\phi(s, a)$	Feature vector	Known functions of (s, a), dimension K
$u(s, a; \theta)$	Flow utility	$\theta^\top \phi(s, a)$ (linear-in-parameters)
$P(s' \mid s, a)$	Transition kernel	Known or estimated from data
$Q(s, a)$	Choice-specific value	$u(s, a; \theta) + \beta \sum_{s'} P(s'\lvert s, a)\, V(s')$
$V(s)$	Integrated value	$\sigma \log \sum_a \exp(Q(s, a)/\sigma)$
$\pi(a \mid s)$	Choice probability	$\exp\big((Q(s, a) - V(s))/\sigma\big)$
τ	Trajectory	$(s_0, a_0, s_1, a_1, \ldots)$ from one agent
$\mathcal{D}$	Panel data	Collection of N observed trajectories

Symbol	Name	Definition
$\Psi(s)$	Sieve basis	Row vector of K basis functions evaluated at s
$\alpha \in \mathbb{R}^K$	Basis coefficients	Weights on the sieve basis
$\omega > 0$	Penalty weight	Controls strength of Bellman enforcement
K	Basis dimension	Number of sieve functions ($K \ll \lvert\mathcal{S}\rvert$)

The sieve approximation replaces $V(s)$ with

$$V(s; \alpha) = \Psi(s)^\top \alpha, \tag{1}$$

where $\Psi(s) = (\psi_1(s), \ldots, \psi_K(s))^\top$ is a deterministic basis. For Fourier bases, $\Psi(s) = (1, \cos(\pi\tilde{s}), \sin(\pi\tilde{s}), \cos(2\pi\tilde{s}), \ldots)$ with $\tilde{s} \in [0, 1]$. Substituting into the Bellman equation gives approximate action values $Q(s, a; \theta, \alpha) = \theta^\top \phi(s, a) + \beta\,(P_a\Psi)^\top \alpha$, where $P_a\Psi$ is precomputed once per action.

The penalized criterion jointly maximizes the log-likelihood and penalizes Bellman violations:

$$\max_{\theta,\alpha} \sum_{i=1}^{N} \log \pi(a_i \mid s_i; \theta, \alpha) \;-\; \omega \sum_{s} \big(V(s;\alpha) - T_\sigma(V;\theta)(s)\big)^2. \tag{2}$$

3 Core Results

Theorem 1 (Sieve Consistency). Under regularity conditions P1–P7 of Luo and Sang (2024), as $\omega = \omega_n \to \infty$ with n, the sieve approximation converges uniformly to the exact equilibrium mapping: $\sup_{\theta\in\Theta} \|\hat{p}(\theta) - p^*(\theta)\|_2 = o_P(1)$. The penalty forces the approximate value function toward the Bellman fixed point. Once the penalty is strong enough, the data likelihood drives the structural parameter estimates. The critical rate condition is $\omega_n/n^2 \to \infty$: the penalty must grow faster than the statistical error, but not so fast that the sieve coefficients α are over-constrained before the data has shaped them.

Theorem 3 (Asymptotic Normality and Efficiency). If $\omega_n/n^2 \to \infty$ and regularity conditions P1–P9 hold, the SEES estimator satisfies

$$\sqrt{n}(\tilde{\theta}_n - \theta_0) \xrightarrow{d} N(0, \Sigma), \qquad \Sigma = -V_{\theta_0}^{-1},$$

attaining the Cramér-Rao lower bound. SEES is asymptotically efficient despite never solving the Bellman equation.

Corollary 3.1 (Schur Complement SEs). The joint Hessian over (θ, α) has a block structure. The marginal Hessian for θ is the Schur complement

$$\tilde{H}_\theta = H_{\theta\theta} - H_{\theta\alpha}\, H_{\alpha\alpha}^{-1}\, H_{\alpha\theta}. \tag{3}$$

Standard errors from $(-\tilde{H}_\theta)^{-1}$ marginalize out the basis coefficients and give the correct Fisher information for the structural parameters. Unlike NNES, which achieves a block-diagonal information matrix via the zero Jacobian property, SEES requires the full Schur complement because the off-diagonal blocks $H_{\theta\alpha}$ are generally nonzero.

Relationship to MPEC. When $K = |\mathcal{S}|$ and $\omega \to \infty$, the sieve expansion becomes exact and the penalty enforces $V = T_\sigma(V;\theta)$ as a hard constraint. In this limit, SEES is the MPEC estimator of Su and Judd (2012), which jointly optimizes (θ, V) subject to the Bellman equation. SEES generalizes MPEC by allowing $K \ll |\mathcal{S}|$, reducing the optimization from $d_\theta + |\mathcal{S}|$ unknowns to $d_\theta + K$ unknowns at the cost of a controlled approximation error.

Projection bias. If the value function has sharp discontinuities that the basis cannot represent, the resulting projection bias contaminates $\hat{\theta}$. The paper recommends increasing K until parameter estimates stabilize. Smooth value functions are well-approximated by Fourier or polynomial bases at moderate K. In the multi-component bus engine, the value function is a smooth function of summed mileage, so a Fourier basis with $K = 8$ terms captures it well.

4 Estimation Algorithm

Algorithm 1: SEES: Joint Algorithm (Luo and Sang, 2024)

Input: Panel $\{(s_i, a_i)\}_{i=1}^N$, features $\phi(s, a)$, transitions $P(s'|s, a)$, discount β, scale σ, basis type, dimension K, penalty ω

`// Step 1: Build sieve basis`
$\Psi \leftarrow$ basis matrix of shape $(|\mathcal{S}|, K)$
Normalize states to $[0, 1]$ (Fourier) or $[-1, 1]$ (polynomial)

`// Step 2: Precompute continuation values`
for *each action* a **do**
  $\mathbb{E}_a[\Psi] \leftarrow P_a \cdot \Psi$ `// shape` $(|\mathcal{S}|, K)$`, one matrix multiply`
end

`// Step 3: Define penalized criterion (Eq. 2)`
$x = [\theta, \alpha] \in \mathbb{R}^{d_\theta + K}$
$f(x) = \sum_i \log \pi(a_i|s_i; \theta, \alpha) - \omega \sum_s (V(s; \alpha) - T_\sigma(V; \theta)(s))^2$

`// Step 4: Joint L-BFGS-B optimization`
$\hat{x} \leftarrow \arg\max_x f(x)$ `// single call, autodiff gradient`
$\hat{\theta}, \hat{\alpha} \leftarrow \hat{x}[: d_\theta],\ \hat{x}[d_\theta :]$

`// Step 5: Standard errors via Schur complement (Eq. 3)`
$H \leftarrow$ numerical Hessian of f at $\hat{x}$
$\tilde{H}_\theta \leftarrow H_{\theta\theta} - H_{\theta\alpha} H_{\alpha\alpha}^{-1} H_{\alpha\theta}$
$\mathrm{SE}(\hat{\theta}) \leftarrow \sqrt{\mathrm{diag}(-\tilde{H}_\theta^{-1})}$

Output: $\hat{\theta}$, V^*, π^*, $\mathrm{SE}(\hat{\theta})$, log-likelihood

Implementation. The entire estimation is a single L-BFGS-B call over the $(d_\theta + K)$-dimensional vector. JAX autodiff computes the gradient through the penalized log-likelihood, including the softmax policy and the soft Bellman operator. The precomputed $\mathbb{E}_a[\Psi] = P_a \Psi$ matrices make continuation value evaluation a matrix-vector multiply, avoiding any inner loop or fixed-point iteration. Bounds on α (default $[-50, 50]$) prevent the basis coefficients from diverging during optimization.

Smoothing parameter selection. Luo and Sang (2024) recommend starting with a moderate ω and multiplying by a factor L until the confidence intervals from successive values of ω overlap by more than a threshold. In practice, values between 1 and 100 work well. At low ω the data shapes the value function freely, producing a good starting point. At high ω the Bellman constraint dominates, pushing the estimator toward MLE. The paper proves that any ω satisfying $\omega_n/n^2 \to \infty$ yields an asymptotically efficient estimator.

5 Results

Table 2: 2-component bus engine (900 states, $\beta = 0.9999$, 10,000 obs). SEES uses a Fourier basis with $K = 8$.

	True	NFXP	SEES
$replacement_cost$	3.0000	2.9274	3.0345
SE	—	0.1115	0.0946
$operating_cost$	0.0010	-0.0477	0.0925
SE	—	0.0460	0.0306
$quadratic_cost$	0.0005	0.0230	-0.0390
SE	—	0.0199	0.0238
Log-lik	—	−1948.7	−1949.0
Time (s)	—	294.1	2.3

We run SEES and NFXP on a 2-component bus engine with 900 total states (30 mileage bins per component), discount $\beta = 0.9999$, and 10,000 observations. SEES uses a Fourier basis with $K = 8$ terms. This environment tests the scaling argument from Luo and Sang (2024): with $|\mathcal{S}| = 900$, NFXP must solve a 900×900 Bellman fixed point at each parameter guess, while SEES optimizes over $d_\theta + 8 = 11$ unknowns regardless of $|\mathcal{S}|$.

SEES achieves a log-likelihood within 0.36 nats of NFXP while running 127× faster (2.3s vs 294.1s). The speedup comes entirely from eliminating the inner fixed-point iteration: at $\beta = 0.9999$, the Bellman contraction rate is $(1 - \beta)$, requiring tens of thousands of value iteration steps per NFXP inner call. SEES bypasses this entirely.

Both estimators agree closely on the replacement cost parameter. The operating and quadratic cost parameters show larger discrepancies because their effects on the likelihood surface are small relative to replacement cost, making them harder to identify precisely. The key observation is that the log-likelihoods are nearly identical: SEES finds the same statistical optimum as NFXP despite optimizing over only 11 unknowns ($d_\theta + K = 3 + 8$) instead of solving a 900×900 fixed point.

Behavioral cloning achieves 99.8% policy accuracy (LL $= -1946.4$), confirming that the data are informative but that BC cannot recover structural parameters or produce standard errors.

SEES occupies a middle ground between MPEC and NNES. The basis functions are deterministic and require no training, making the optimization smooth and well-conditioned. The trade-off is projection bias when V is not well-represented by the basis. NNES (Nguyen, 2025) avoids this with a universal approximator and achieves valid SEs via Neyman orthogonality, at the cost of alternating optimization and sensitivity to initialization.

Code. The SEES estimator is implemented in `src/econirl/estimation/sees.py`. The companion script `sees_run.py` generates Table 2. The key implementation details map directly to Algorithm 1: `_build_basis()` constructs Ψ (Step 1), the `penalized_ll()` closure implements Equation 2 (Step 3), and the Schur complement in Equation 3 is computed from the numerical Hessian (Step 5). To regenerate:

```
python papers/econirl_package/primers/sees/sees_run.py
```

References

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